

MAQAM: A Richly Annotated Corpus for Quranic Arabic Morphological Analysis and Disambiguation

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Abstract

Quranic Arabic poses important challenges for Arabic NLP due to its distinctive morphology, syntax, and orthography. We present **MAQAM**, a richly annotated Morphosyntactic Arabic Quranic Analysis and Modeling resource covering the entire Quran, providing 22 lexical, inflectional, and clitic features for every word. The resource is constructed through a semi-automatic pipeline combining feature enrichment, morphological synchronization, and expert manual annotation and validation. We further evaluate Quranic morphosyntactic disambiguation under a realistic setting where the input may be represented in either the Uthmani or Imla'i script, and investigate cross-domain transfer between Quranic Arabic and MSA. Our results show that MSA and Quranic resources are complementary for Quranic disambiguation, with the best performance obtained by combining MSA training data, Quran-specific analyzers, and both script representations, whereas for MSA, only integrating Quranic resources provide modest gains. We make **MAQAM** publicly available.¹

1 Introduction

The Quran is central to both Islamic studies and Arabic linguistics and serves as an important benchmark for Arabic natural language processing (NLP). However, Quranic Arabic differs substantially from Modern Standard Arabic (MSA) and contemporary dialects in its morphology, syntax, and orthography, limiting the effectiveness of NLP tools developed primarily for MSA.

While Arabic NLP has primarily focused on MSA and, more recently, Arabic dialects, Classical Arabic has received comparatively less attention. Moreover, research on Quranic Arabic and MSA has largely progressed independently, with limited investigation of how resources and models transfer

Orthography	Uthmani	وَإِذْ قَالَتِ الْمَلَائِكَةُ يَا مَرْيَمُ
	Imla'i	وَإِذْ قَالَتِ الْمَلَائِكَةُ يَا مَرْيَمُ
Annotated Resources	Quran Morph	اسم, مَلَكٌ
	Quranic Arabic Corpus	DET PREFIX +JI N STEM POS:N LEM:مَلَكٌ ROOT:ملك MP NOM
	MASAQ	JI, DET, Prefix, DEF_ART, the-angels مَلَكٌ, NOUN_CONCRETE, Stem, DECLN, AGNT, NOT_CONSTRUCT, NOMINATIVE, DHAMMA, the-angels ة, NSUFF_FEM_SG, Suffix, the-angels
	MAQAM (Our)	lex:مَلَكٌ pos:noun gloss:angel root:ملك prc3:0 prc2:0 prc1:0 prc0:AI_det enc0:0 enc1:0 per:na gen:m form_gen:f num:p form_num:s asp:na vox:na mod:na stt:d cas:n stem:مَلَكَ rat:r

Figure 1: Annotation comparison for الْمَلَائِكَةُ *AlmAlAykāh* ‘when the angels said, O Mary! (3:42)’.

between the two varieties. A better understanding of their linguistic and computational relationship can facilitate the transfer of linguistic resources and NLP methodologies across both domains. This work investigates that relationship through morphological and morphosyntactic analysis, assessing the transferability of resources and tools between Quranic Arabic and MSA.

Over the years, numerous studies have explored Quranic NLP applications, including question answering (Saadaoui et al., 2024), topic modeling (Zafar et al., 2024), Lemmatization (Saeed and Habash, 2025) and diacritization (Cherradi and El Mahajer, 2025). Complementing these downstream applications, several Quranic NLP shared tasks have emerged to address a variety of linguistic and semantic challenges (Boucekif et al., 2025; Mubarak et al., 2025; El Kheir et al., 2025; Basem et al., 2024; Malhas et al., 2023, 2022). These efforts underscore the need for comprehensive and richly annotated Quranic linguistic resources.

¹github.com/CAMeL-Lab/MAQAM-Quranic-Corpus.

Figure 1 compares the annotation of a Quranic word across existing resources and the Uthmani and Imla'i scripts. The Quranic Arabic Corpus (QAC) (Dukes and Habash, 2010) and Quran-Morph (Akra et al., 2025) use Uthmani, whereas MASAQ (Sawalha et al., 2025) uses Imla'i. Red highlights mark orthographic differences, while other colors indicate shared features and gray marks resource-specific features. The example reveals substantial variation in segmentation, annotation granularity, and feature coverage, often requiring information from multiple morphemes or fields to obtain a complete representation, with some annotations unavailable throughout the Quran.

In contrast, our proposed corpus consolidates all lexical, inflectional, and morphosyntactic information for each word into a single, consistently annotated record, eliminating the need to reconstruct annotations from multiple morphemes while providing richer and more comprehensive feature coverage. Furthermore, while modern Arabic NLP disambiguation tools, such as the CAMEL Tools disambiguator (Obeid et al., 2020; Inoue et al., 2022), perform well on MSA, they are not specifically designed for Quranic Arabic. We therefore use the CAMEL Tools disambiguator as a strong baseline and develop enhancements tailored to the linguistic characteristics of the Quran.

Our main contributions are as follows:

- **A New Quranic Morphological Resource.** We present **MAQAM** (Morphosyntactic Arabic Quranic Analysis and Modeling Resource) a richly annotated morphosyntactic resource for the entire Quran. We make this resource publicly available to support Arabic NLP research.^{1,2}
- **Quranic Morphosyntactic Disambiguation.** To the best of our knowledge, we present the first benchmark for Quranic morphosyntactic disambiguation, under a realistic setting supporting both Uthmani and Imla'i scripts.
- **Cross-Domain Evaluation.** To the best of our knowledge, we present the first cross-domain evaluation of morphosyntactic disambiguation between Quranic Arabic and MSA.

The remainder of this paper presents the linguistic background and related work, describes the proposed resource and methodology, and reports experimental results and error analysis.

2 Linguistic Background & Challenges

Arabic comprises several varieties, including Classical Arabic, MSA, and regional dialects. Differences between Classical Arabic and MSA in vocabulary, morphology, syntax, and orthography limit the transferability of MSA-based NLP tools to Quranic Arabic. Moreover, the Quran is represented in both *Uthmani* and *Imla'i* scripts,³ with only 64.6% token overlap in their diacritized forms and 85.2% after removing diacritics, indicating that approximately 15% of the differences extend beyond diacritic variation.

From a morphosyntactic perspective, the differences between Quranic Arabic and MSA can be broadly categorized into orthographic, lexical, and inflectional variation. Orthographic variation concerns differences in surface word forms and spelling conventions. For example, the Uthmani form *أُمْرَأَتُ عِمْرَانَ* *Äm.raÄatu ñim.raána*⁴ ‘the wife of Imran (3:35)’ corresponds to the Imla'i form *أُمْرَأَتُ عِمْرَانَ* *Äm.raÄatu ñim.raAna*, while MSA typically uses *أُمْرَأَةُ عِمْرَانَ* *Äm.raÄahu ñim.raAna*. Such variation complicates lexicon lookup, morphological analysis, and resource alignment across Arabic varieties. Lexical variation is also evident in the Quranic lemma inventory. As will be illustrated later, manual word-by-word lemma annotation by a linguistic expert revealed 343 Quranic lemmas that are not covered by the MSA morphological database used in this study, as discussed later in the paper. These out-of-vocabulary lemmas span several categories, including proper nouns such as *هَارُوتَ* *haAruwta* ‘Harut (2:102)’, Quranic disjoint letters, abbreviations such as *طسم* *Tsm* (26:1), and rare lexical items and verbal nouns such as *تَتْبِيْبٌ* *tat.biybī* ‘ruin (11:101)’.

Inflectional variation further distinguishes Quranic Arabic from MSA. Quranic Arabic frequently employs emphatic constructions that are less common in contemporary MSA usage, such as the co-occurrence of *inna* and *emphatic lam*, as in *إِنَّ فِي ذَلِكَ لَذِكْرَى* *Äin a fiy ðaálaka laðik.raýá* ‘Surely in this is a reminder (39:21)’. Similarly, the emphatic nun (energetic mood) occurs in forms such as *لَيَكُونَنَّ* *layakuwnnan~a* ‘surely, they would be (35:42)’, while MSA typically

²The used Quran recitation version is Hafs ‘an ‘Asim.

³Data source: <https://tanzil.net>.

⁴Arabic in HSB Romanization (Habash et al., 2007).

avoids its use. Additional variation appears in the realization of clitics. For example, the final pronominal clitic is omitted in the word *لِيَعْبُدُونِ* *liyaṣ.buduwni* ‘that they worship Me’ in the verse *وَمَا خَلَقْتُ الْجِنَّ وَالْإِنْسَ إِلَّا لِيَعْبُدُونِ* *wamaA xalaq.tu Āl.jin~a waĀl.Āinsa Āil aA liyaṣ.buduwni* ‘I did not create jinn and humans except to worship Me. (51:56)’, whereas the corresponding MSA form would normally be *لِيَعْبُدُونِي* *liyaṣ.buduwniy*.

These orthographic, lexical, and inflectional differences collectively distinguish Quranic Arabic from MSA and contribute to the limited transferability of existing MSA-oriented NLP resources and models. As a result, morphological analyzers, disambiguation systems, and language models trained primarily on MSA often struggle to accurately process Quranic text, highlighting the need for dedicated Quranic Arabic resources and adaptation techniques.

3 Related Work

As previously mentioned, several efforts have annotated Quranic data at different linguistic levels, including morphology, syntax, and semantics (Dukes and Habash, 2010; Sawalha et al., 2025; Akra et al., 2025; Faqihi et al., 2026). However, these resources differ in their annotation schemes, objectives, and coverage, and some exhibit inconsistencies in annotation granularity across the Quran. These differences are discussed below.

The **Quranic Arabic Corpus (QAC)** is one of the earliest and most influential morphologically annotated resources for Classical Arabic. It was developed through a combination of automatic analysis using an adapted Buckwalter Arabic Morphological Analyzer (BAMA) and extensive manual refinement (Buckwalter, 2004) and the features are annotated at the morpheme level rather than the token level. While the corpus introduces rich morphological features that make it a foundational resource for Quranic Arabic research, it suffers from inconsistencies in feature assignment, where some words are annotated with more or fewer features than others.

The **QuranMorph Corpus** (Akra et al., 2025) is an annotated Quranic dataset providing lemma and part-of-speech annotations for each token by three linguists using the Tawseem Web Annotation tool. Lemmas are sourced from the Qabas lexicographic database (Jarrar and Hammouda, 2024),

	MAQAM	QAC	QM	MASAQ	QAMAR
<i>Orthographic Style</i>					
Uthmani	✓	✓	✓		✓
Imla’i	✓			✓	✓
<i>Lexical Features</i>					
Gloss	✓			✓	
Lex	✓	✓	✓		✓
Rationality	✓				
Root	✓	✓			✓
Stem	✓	✓		✓	✓
<i>Morphological Features</i>					
Aspect	✓	✓			
Case	✓			✓	
Gender	✓	✓			
Number	✓	✓			
Mood	✓			✓	
Person	✓				
POS	✓	✓	✓	✓	✓
State	✓			✓	
Voice	✓	✓			
Clitics	✓	✓		✓	

Table 1: Comparison of available scripts and morphological features across Quranic Arabic resources. QM (QuranMorph), QAC (Quranic Arabic Corpus).

which integrates 110 lexicons and corpora totaling over two million tokens, and POS tags follow the 40-category SAMA/Qabas tagset (Maamouri et al., 2010b; Jarrar and Hammouda, 2024). However, the corpus does not include morphological, syntactic, or semantic annotations.

The **MASAQ Dataset** (Sawalha et al., 2025) is a morphosyntactically annotated corpus covering the entire Quran. Built on the verified Tanzil text, it provides detailed morphological and syntactic annotations for each word on the morpheme level not the token level, produced and validated by expert linguists. The resulting resource offers rich grammatical information of Quranic Arabic but still missing other morphosyntactic features.

The **Quranic Arabic Morphological Analysis Resource (QAMAR)** corpus (Faqihi et al., 2026) is a manually verified morphological resource covering the entire Quran in both Uthmani and Imla’i forms. Developed using SAFAR-generated analyses followed by expert validation, it provides stems, lemmas, roots, and POS annotations while preserving Quranic orthography and providing normalized MSA representations.

As shown in Table 1, current Quranic resources

Orthography	Imla'i	وَالَّذِينَ	وَعَمِلُوا	آمَنُوا	وَالصَّالِحَاتِ
		waAl~ađiyna	waçamiluwA	ĀmanuwA	AlS~aAliHaAti
	Uthmani	وَالَّذِينَ	وَعَمِلُوا	ءَامَنُوا	الصَّالِحَاتِ
Lexical Features		Ala~ađiy	çamil	Āman	SaAliHah
	gloss	whom	work	believe	good deed
	lex	الَّذِي	عَمِلَ	آمَنَ	صَالِحَةٌ
Inflectional Features	rat	na	na	na	i
	root	NTWS	أ.م.ن	ع.م.ل	ص.ل.ح
		NTWS	Ā.m.n	ç.m.l	S.l.H
	stem	الَّذِينَ	آمَنَ	عَمِلَ	صَالِحَ
		Ala~ađay.ni	Āman	çamil	SaAliH
Clitic Features	asp	na	p	p	na
	cas	n	na	na	a
	form_gen	m	m	m	f
	form_num	s	p	p	p
	gen	m	m	m	f
	num	p	p	p	p
	mod	na	i	i	na
	per	na	3	3	na
	pos	pron_rel	verb	verb	noun
	stt	d	na	na	d
	vox	na	a	a	na
	prc3	0	0	0	0
	prc2	wa_conj	0	wa_conj	0
	prc1	0	0	0	0
	prc0	0	0	0	Al_det
	enc0	0	0	0	0
	enc1	0	0	0	0

Figure 2: Sample entries from MAQAM corpus with their associated lexical, inflectional, and clitic features: ‘And those who believe and do good deeds (2:82).’

do not provide complete morphosyntactic annotation of every word in the Quran, with several linguistic features either missing, inconsistently represented, or unavailable. To fill this gap, we introduce **MAQAM**, a comprehensive Quranic corpus providing a unified set of morphosyntactic annotations for the whole Quran, supporting both fine-grained linguistic analysis and Quranic Arabic NLP.

4 The MAQAM Corpus Design

4.1 Word Entries

Each entry in MAQAM corresponds to a single Quranic token and includes both *Uthmani* and *Imla'i* representations together with unified lexical, inflectional, and clitic morphosyntactic features. Figure 2 summarizes the annotation schema.

Dataset	# Surahs	# Verses	# Words
Q-All	114	6,236	77,429
Q-Train	101	5,605	71,801
Q-Test	13	631	5,628

Table 2: Corpus splits for MAQAM.

4.2 Corpus Split

The corpus is organized into three splits: **Q-All**, **Q-Train**, and **Q-Test**. Table 2 summarizes their statistics in terms of surahs, verses, and words. Q-All covers the entire Quran and is partitioned into Q-Train and Q-Test. Q-Test comprises the 13 surahs located in the middle of the Quran, spanning Surahs 45–57, while Q-Train contains the remaining 101 surahs. All words in Q-All are manually annotated for lemma, part of speech, and gloss. Q-Train inherits these lexical annotations, while Q-Test is additionally fully manually validated for all morphosyntactic features for each word in it. We use the subscripted versions Q-Train_i and Q-Train_u to refer to the Imla'i vs Uthmani sub-corpus, respectively. When no subscript is used, both versions are included. The annotation and validation procedure is described in Section 5.3.

5 Corpus Construction Pipeline

Our corpus construction approach relies on four resources that are integrated through a synchronization process that integrates their complementary content. We present each of these resources next followed by the synchronization step.

5.1 Imla'i-Uthmani Spelling Alignment

We obtain the *Imla'i* and *Uthmani* Quranic scripts from the Tanzil website. Although both represent the same Quranic text, in some cases they differ in tokenization where a single token in one script corresponds to multiple tokens in the other. We therefore perform a token-level alignment between the two representations using the Character Edit Distance (CED) word alignment tool (Khalifa et al.), which is later used to map the morphosyntactic annotations from the QAC to the corresponding tokens. The final alignment achieves **100% accuracy**, ensuring an exact mapping between the tokens and their corresponding analyses for the whole Quran.

5.2 QAC Feature Extraction

After obtaining the alignment between the *Imla'i* and *Uthmani* representations, we extract the corresponding annotations from the QAC corpus for each aligned word. Since QAC follows a different annotation scheme, we convert its analyses into unified feature-value pairs and then enrich them using an extended version of the Arabic Extended Morphological Analysis and Disambiguation Tagset system (EMAD) (Kallas et al., 2024), a bridge tagset for unifying Arabic morphological annotation schemes. Our extension to EMAD adds support for enclitics, orthographic gender and number, emphatic mood, and enhanced feature mappings. The pipeline preprocesses QAC annotations, converts them into the EMAD representation, and applies consistency checks to remove invalid analyses. Due to morphological ambiguity, a word may produce multiple valid feature-value analyses, which are resolved during synchronization.

5.3 Manual Annotation

As mentioned earlier, the QAC has several limitations for our corpus construction: lemmas are represented in *Uthmani* orthography, some words lack lemma annotations, glosses are not provided, and POS tags are often underspecified (e.g., N and ADJ correspond to multiple fine-grained categories). To address these limitations, every Quranic word was manually annotated with its lemma, fine-grained POS, English gloss, root, and rationality (Alkuhlani and Habash, 2012; Zaina et al., 2026). The rest of the morphosyntactic features were automatically extended (see Section 5.5). However, every morphosyntactic feature in the Q-Test split was further manually validated,⁵ producing a high-quality benchmark later used to evaluate downstream morphosyntactic disambiguation.

Our annotation team consisted of two expert native Arabic speakers with extensive experience in Arabic computational linguistics annotation: an Egyptian woman with a BA in English Language and Literature and working on a MS in Linguistics, and a Lebanese man with a BA and an MS in Philosophy. The former completed the lemma, POS, and gloss annotations for Q-All, while both annotators contributed to the rationality and root annotations, and the full Q-Test split validation. For inter-annotator agreement (IAA) on the 14 mor-

Level	Script	R13	S31	CM
Tokens	Uthmani	5.8	5.4	3.2
	Imla'i	2.1	1.8	0.8
Types	Uthmani	10.8	9.9	5.9
	Imla'i	6.6	5.7	2.7

Table 3: MAQAM OOV rates (%) under different orthographic representations and morphological databases.

phosyntactic features,⁶ we used two Q-Test surahs comprising 702 tokens and 9,828 annotation cells (702×14). The IAA was **99.1%** for Surah 51 and **98.7%** for Surah 54, with an average IAA of **98.9%**. These manual feature annotations were then used to guide the automatic synchronization of the remaining morphosyntactic features.

The annotation process was time-intensive and required considerable care and research, including consultation of linguistic references and Quranic resources. Quranic Arabic has distinctive lexical, morphological, and syntactic properties, and its interpretation requires specialized, context-sensitive analysis informed by knowledge of Classical Arabic vocabulary and Quran-specific references. The two annotators and the lead authors met regularly to discuss and resolve manual annotation decisions. Nevertheless, we recognize that some decisions may admit alternative interpretations, including interpretations that we did not anticipate. We acknowledge this as an inherent limitation of annotating Quranic Arabic and welcome future extensions and corrections from the broader research community.

5.4 Arabic Morphological Analyzer

The upcoming synchronization process relies on a morphological analyzer to generate candidate analyses for each Quranic word. We compare three MSA morphological databases: Calima-MSA-R13 (R13) (Buckwalter, 2002), Calima-MSA-S31 (S31) (Maamouri et al., 2010a; Taji et al., 2018), and CAMeL Morph (CM) (Khairallah et al., 2024).

Table 3 reports the out-of-vocabulary (OOV) rates of the three databases under both the *Uthmani* and *Imla'i* script representations. Among them, CAMeL Morph consistently achieves the lowest OOV rates at both the token and type levels. We therefore adopt CAMeL Morph with the *Imla'i* representation in the synchronization process.

⁵Validated Features: *lex, pos, prc3, prc2, prc1, prc0, per, asp, vox, mod, gen, num, stt, cas, enc0, and enc1*.

⁶We exclude lemma and POS from IAA because the annotators jointly validated them; and we exclude the *form_gen* and *form_num* because they are deterministic.

5.5 Synchronization

Finally, the synchronization process integrates the resources described above to produce a unified morphosyntactic annotation for every Quranic word. On one side, we use the enriched QAC annotations together with the manually annotated lemma, POS, and gloss. On the other side, we generate candidate analyses using CAMEL Morph using the CAMEL Tools disambiguation framework.

Synchronization is formulated as selecting the CAMEL Morph analysis that best matches one of the enriched Quranic annotations. Each candidate analysis is compared against the enriched annotation using the shared features *diac*, *lex*, *POS*, *per*, *asp*, *vox*, *mod*, *gen*, *num*, *form_gen*, *form_num*, *stt*, *cas*, *prc0-prc3* and *enc0-enc1*. A matching score is computed based on feature agreement, and the highest-scoring analysis is selected as the synchronized annotation. When multiple analyses obtain the same highest matching score, ties are resolved using the CAMEL Tools disambiguator, which reranks the analyses generated by CAMEL Morph according to how well they match the predicted features.

The synchronization procedure was iteratively refined using the manually validated Q-Test benchmark, allowing synchronization errors to be identified and corrected. The full process achieves an average accuracy of 98.3% on Q-Test. A detailed breakdown by morphosyntactic feature and vocabulary status (all, INV, and OOV words) is provided in Appendix B. This evaluation covers the features produced through automatic synchronization while *lex*, *POS*, *gloss*, *rat*, and *root* are excluded because they were manually annotated across the entire corpus and not automatically generated.

6 Morphosyntactic Disambiguation

Following the construction of the synchronized data, we investigate Quranic morphosyntactic disambiguation with the CAMEL Tools unifactored MSA disambiguator as a baseline. We examine the relationship between MSA and Quranic Arabic by varying the training data and morphological analyzer and evaluating bidirectional transfer. For MSA, we use the Penn Arabic Treebank (PATB) (Maamouri et al., 2004), on which the baseline is trained, while for Quranic Arabic, we use MAQAM. This assesses whether Quran-specific resources improve Quranic disambiguation while preserving MSA performance.

The unifactored disambiguator formulates morphosyntactic analysis as a multiclass classification task, where each class represents a complete morphosyntactic analysis set defined by 14 features: *POS*, *prc0-prc3*, *per*, *asp*, *vox*, *mod*, *form_gen*, *form_num*, *stt*, *cas*, and *enc0*. To illustrate the diversity of morphosyntactic analyses covered by each training configuration, we report the number of unique class labels in each dataset. Q-Train contains 1,939 classes, Q-All 1,982, Q-Test 695, and PATB 2,016. Combining Quranic and PATB data increases the unique label count to 3,213 classes for PATB+Q-Train and 3,269 classes for PATB+Q-All.

During inference, the model predicts the most likely class for each word in context, and candidate analyses from a morphological analyzer are ranked according to their agreement with the predicted class. The highest-ranked analysis in the top ranked set is selected, with ties resolved using *POS_lex_logprob* followed by *lex_logprob*.

Accordingly, the proposed morphological disambiguation framework relies on three key elements:

Disambiguator Training Data We fine-tune the disambiguator using the PATB training set, representing Modern Standard Arabic (MSA); the Quranic training split (Q-Train); the Quranic dataset (Q-All); and the datasets PATB + Q-Train and PATB + Q-All. For configurations containing Quranic data, we train using the Imla’i script, the Uthmani script, or both. These configurations allow us to examine the contributions of MSA and Quranic Arabic data and the effect of script representation on disambiguation performance.

Morphological Analyzer Analyses come from three sources: (1) MSA databases (R13, S31, and CAMEL Morph), (2) Quran-specific analyzers derived from Q-Train and Q-All, and (3) The combination of MSA and Quranic analyzers. Each Quran-specific analyzer is a lookup table mapping an undiacritized word to corpus-observed analyses, constructed from the Imla’i representation, the Uthmani representation, or both. When combined, in-domain analyses are prioritized: Quran-specific analyses for Quranic evaluation and MSA analyses for MSA evaluation.

Evaluation Corpus Models are evaluated on Q-Test using both Imla’i and Uthmani scripts as input and the PATB test set to measure in-domain performance and cross-domain transfer between Quranic Arabic and MSA.

	Training Data	Analyzer Database	Lexical	Inflectional		Clitics		All	
			Exact	Macro Avg	Exact	Macro Avg	Exact	Macro Avg	Exact
(a)	PATB	R-13	74.0	89.7	51.5	98.7	94.0	91.8	44.7
	PATB	S31	78.1	92.3	61.0	99.0	95.4	93.7	54.0
	PATB	CM	82.7	93.6	68.5	98.8	93.8	94.7	61.3
(b)	Q-Train	CM	85.7	95.6	75.4	99.2	95.7	96.2	67.1
	Q-Train	Q-Train	82.8	90.6	67.0	98.9	93.9	92.9	64.7
	Q-Train	CM+Q-Train	92.3	95.9	76.8	99.2	96.0	96.8	72.9
(c)	PATB+Q-Train	CM	86.0	96.1	76.6	99.3	95.9	96.6	68.1
	PATB+Q-Train	Q-Train	82.9	91.0	67.4	98.9	94.0	93.2	64.6
	PATB+Q-Train	CM+Q-Train	92.7	96.5	78.0	99.3	96.2	97.2	73.9
(d)	PATB+Q-All	CM+Q-All	98.7	97.9	84.4	99.3	96.4	98.4	81.0
(e)	PATB+Q-Train _i	CM+Q-Train _i	92.4	96.4	77.7	99.3	96.2	97.2	73.6

Table 4: Morphosyntactic tagging performance on the **Imla’i Q-Test_i** set across different fine-tuning and morphological analyzer configurations.

	Training Data	Analyzer Database	Lexical	Inflectional		Clitics		All	
			Exact	Macro Avg	Exact	Macro Avg	Exact	Macro Avg	Exact
(a)	PATB	R-13	67.0	85.0	46.8	98.2	91.6	88.4	40.3
	PATB	S31	71.3	88.6	55.6	98.4	92.9	90.9	49.0
	PATB	CM	75.5	90.1	62.6	98.1	91.1	91.9	55.8
(b)	Q-Train	CM	78.5	92.3	69.2	98.6	93.7	93.7	60.8
	Q-Train	Q-Train	82.6	90.5	66.8	98.8	93.7	92.8	64.4
	Q-Train	CM+Q-Train	91.0	95.4	75.8	99.1	95.5	96.4	71.6
(c)	PATB+Q-Train	CM	78.7	92.8	70.2	98.7	93.7	94.0	61.7
	PATB+Q-Train	Q-Train	82.7	90.9	67.2	98.9	93.8	93.1	64.3
	PATB+Q-Train	CM+Q-Train	91.3	95.9	76.7	99.2	95.7	96.8	72.5
(d)	PATB+Q-All	CM+Q-All	98.5	97.8	84.3	99.3	96.4	98.4	80.8
(e)	PATB+Q-Train _u	CM+Q-Train _u	91.1	95.8	76.6	99.2	95.7	96.7	72.3

Table 5: Morphosyntactic tagging performance on the **Uthmani Q-Test_u** set across different fine-tuning and morphological analyzer configurations.

7 Experiments and Results

7.1 Experimental Setup

Fine-tuning Setup Following the fine-tuning setup of (Inoue et al., 2022), we fine-tuned CAMEL-Lab/bert-base-arabic-camelbert-msa as a token-classification model on a single NVIDIA A100 GPU. Each word was assigned a composite label representing 14 morphosyntactic features, with supervision applied only to its first subword. Models were trained for 10 epochs using a batch size of 32, a learning rate of 5×10^{-5} , a maximum sequence length of 512 tokens, and a fixed seed of 12,345. The best checkpoint was selected based on validation accuracy evaluated every 500 training steps.

Evaluation Metrics Performance was evaluated using macro-average per-feature accuracy and exact-match (EM) accuracy across four feature

groups: **Lexical** (*lex*), **Inflectional** (*asp*, *cas*, *form_gen*, *num*, *gen*, *mod*, *form_num*, *per*, *POS*, *stt*, *vox*), **Clitic** (*prc0-prc3*, *enc0*, *enc1*), and **Overall** (all 18 features). EM considers a prediction correct only when all features within the corresponding group match the gold analysis.

Disambiguation Configuration Predicted composite labels were used to rank the analyses generated by the morphological analyzer. All systems used the **NOAN_PROP** backoff strategy, which assigns a proper-noun analysis to words that are OOV with respect to the analyzer.

7.2 Results

Tables 4 and 5 present morphosyntactic tagging performance on the Imla’i and Uthmani Q-Test sets under different combinations of disambiguator training data and morphological analysis resources. All Quranic training data and Quran-specific analyzers

	Training	Analyzer	Lexical	Inflectional		Clitics		All	
	Data	Database	Exact	Macro Avg.	Exact	Macro Avg.	Exact	Macro Avg.	Exact
(a)	PATB	CM	93.5	98.5	90.4	99.8	99.2	98.7	86.3
	PATB	Q-All	40.9	85.3	37.8	99.6	97.5	87.6	32.6
	PATB	CM+Q-All	93.6	98.5	90.4	99.8	99.2	98.7	86.3
(b)	PATB+Q-All	CM	93.5	98.5	90.3	99.8	99.2	98.7	86.3
	PATB+Q-All	Q-All	40.8	85.3	37.8	99.6	97.5	87.6	32.6
	PATB+Q-All	CM+Q-All	93.6	98.5	90.4	99.8	99.2	98.7	86.3

Table 6: Morphosyntactic tagging performance on the PATB test set (63,172 words). Results are reported as percentages (%).

in Groups (a)–(d) are constructed from the combined Imla’i and Uthmani representations. Group (e) instead uses only the script corresponding to the evaluation set, allowing us to assess the benefit of dual-script training.

Groups (b) and (c) investigate the contribution of Quranic resources. In group (b), the disambiguator is trained only on Quranic data, with candidate analyses from the CM analyzer, the Q-Train analyzer, or both. CM outperforms Q-Train on all feature groups except lexical prediction, while combining both consistently yields the best performance, demonstrating the complementary strengths of MSA and Quran-specific resources.

Group (c) further combines MSA and Quranic data during disambiguator training while retaining the combined analyzer. This configuration achieves the best non-oracle performance across all feature groups, improving overall macro accuracy and exact match accuracy compared with training only on Quranic data. These results show that MSA and Quranic Arabic are complementary: MSA training data improve contextual disambiguation, while Quran-specific lexical resources improve coverage of Quranic vocabulary.

Finally, groups (d) and (e) provide two reference settings for interpreting the results. Group (d) represents an oracle experiment in which both the disambiguator and the Quran-specific analyzer are constructed using Q-All. Because Q-All includes Q-Test, this setting is not intended as a realistic deployment scenario; rather, it provides an approximate upper bound on the performance achievable when the test data are covered during training. Group (e), in contrast, uses the best-performing non-oracle configuration but is trained only on the Imla’i representation. Compared with the dual-script configuration in group (c), this model performs slightly worse on some feature groups, although the differences are generally small but still,

this difference demonstrates the benefit of jointly supporting both the Imla’i and Uthmani input representations during training. In practice, we therefore recommend the non-oracle dual-script configuration in group (c), since it avoids test-set leakage while allowing newly annotated Quranic data to be incorporated into future versions of the training set.

Table 5 presents the results for the Uthmani representation. The same trends observed for Imla’i hold, although performance is lower. This is likely because Uthmani orthography differs considerably from MSA conventions, limiting the transferability of MSA-trained resources and highlighting the need for Quran-specific data when processing this script.

Table 6 reports the evaluation on the PATB test set. Unlike the Quranic evaluation, adding Quranic training data results in nearly identical MSA performance, with a marginal improvement of 0.1% for lexical feature EM accuracy. All other results remain unchanged, indicating that incorporating Quranic data neither improves nor degrades MSA disambiguation performance, consistent with the nearly identical results reported in the evaluation table. This may be because PATB represents the newswire genre, whereas Quranic resources primarily benefit Classical Arabic in the religious genre.

We assess significance using McNemar’s test (McNemar, 1947) on paired word-level predictions under the overall exact-match metric. We compare the CAMEL Tools morphological-disambiguation baseline with the best-performing non-oracle system in Group (c). The improvements on the Imla’i and Uthmani Q-Test sets are statistically significant ($p < 0.05$). In contrast, the difference between the baseline and the best-performing system on the PATB test set is not statistically significant ($p \geq 0.05$), consistent with the nearly identical performance reported in the evaluation table.

Word	Analysis	Imla’i	Uthmani
OOV	OOV	2.7	3.7
INV	OOV	46.4	40.5
INV	INV	50.9	55.8

Table 7: Distribution of analyzer and disambiguator errors (%) comparing in-vocabulary and out-of-vocabulary patterns.

8 Error Analysis

To better understand the remaining errors, we analyze incorrectly predicted instances under the All Tags exact-match (EM) metric using the best-performing non-oracle system.

Table 7 summarizes this breakdown. For the Imla’i representation, only 2.7% of errors are due to out-of-vocabulary (OOV) words not covered by the analyzer, while 46.4% occur because the correct analysis is absent from the analyzer’s candidate set despite the word being in vocabulary. The remaining 50.9% are true disambiguation errors, where the correct analysis is generated by the analyzer but the disambiguator selects a different candidate. Similar trends are observed for Uthmani, with 3.7%, 40.5%, and 55.8% of errors falling into these three categories, respectively.

We also compare the predictions produced by the Imla’i and Uthmani representations under the All Tags EM metric using the best performing system. The two systems produce the same prediction for the vast majority of words, correctly analyzing 72.1% of the test set and jointly failing on 25.9%. However, the Imla’i representation correctly predicts an additional 1.8% of words that Uthmani misses, whereas Uthmani uniquely predicts only 0.3%. This further demonstrates the consistent advantage of the Imla’i representation for morphosyntactic disambiguation.

The remaining errors are concentrated in a small number of morphosyntactic features rather than being distributed uniformly across the complete feature set. Among the inflectional features, *case* is the largest source of difficulty, accounting for more than a quarter of all errors (27.3%). This is followed by *gender* (12.9%), *state* (12.8%), *form_gen* (11.2%), and *POS* (11.1%), whereas every other inflectional feature individually contributes less than 6%. This distribution suggests that future improvements should primarily target features whose correct values depend strongly on syntactic context or distinctions between closely related analyses.

Clitic errors exhibit an even stronger concentration: *prc2* (41.5%) and *prc1* (33.9%) jointly account for more than three quarters of all clitic errors. By comparison, enclitic errors are less frequent, representing 17.8% for *enc0* and less than 1% for *enc1*. Overall, these findings identify case assignment and proclitic disambiguation as the most important targets for reducing the remaining exact-match errors.

9 Conclusion & Future Work

In this paper, we introduced MAQAM, a comprehensive morphosyntactic resource providing unified token-level annotations for every Quranic word in both the Imla’i and Uthmani scripts. We also presented a synchronization framework and established a benchmark for Quranic morphosyntactic disambiguation. Our experiments demonstrate that MSA and Quranic resources are complementary: MSA data improve contextual disambiguation, while Quran-specific analyzers increase the coverage of valid Quranic analyses. The best non-oracle system significantly outperforms the CAMEL Tools baseline on both script representations, with Imla’i consistently performing better than Uthmani, while the differences on the PATB test set are not statistically significant which may be due to the domain difference between the news-oriented PATB and the newly introduced Quranic corpus.

We will publicly release the MAQAM corpus, synchronization pipeline, trained models, and evaluation code to support further research on Quranic and Classical Arabic NLP. Future work will focus on improving candidate generation and contextual disambiguation, particularly when the correct analysis is missing from the analyzer’s candidate set. We also plan to investigate productive Quranic morphological analyzers that can generalize beyond observed corpus forms, as well as factored and multi-task models that predict individual morphosyntactic features and reduce label sparsity. Additional directions include improving the handling of Uthmani orthography, integrating recent Quranic lemmatization models (Saeed and Habash, 2025) into the pipeline to improve lexical prediction and overall morphosyntactic analysis, extending the evaluation to other Classical Arabic and religious-text genres, and assessing MAQAM’s usefulness for downstream tasks such as diacritization.

Limitations

A key limitation of the current system is its reliance on analyzer-generated candidate analyses. When the correct analysis is absent from the candidate set, the disambiguator cannot recover it, regardless of the quality of the tagging model. Addressing this limitation will require more effective backoff or candidate generation strategies that can produce plausible analyses for unseen or missing forms. In addition, the Quran-specific analyzers used in this work are implemented as lookup tables over observed corpus forms, limiting their ability to generalize to unseen Classical Arabic words or generate productive morphological analyses beyond the Quran. Finally, the unfactored disambiguator models each complete morphosyntactic analysis as a single class label, resulting in a large and sparse label space that may hinder generalization to rare or unseen feature combinations.

Ethics Statement

All manual annotations used to construct the resource were performed by qualified annotators who were compensated at fair market rates for their work.

Because the Quran is the sacred scripture of over 2 billion Muslims, we approached its annotation and processing with care and respect for its religious significance. Throughout the development of this resource, we sought to faithfully preserve the original text while employing a rigorous and transparent scientific methodology. Although the resource underwent extensive automatic validation and expert manual review, some errors may nevertheless remain. Any such errors are solely our responsibility, and we welcome feedback and corrections from the research community to further improve the resource.

The annotations represent linguistic decisions made for research purposes and should not be treated as authoritative interpretations of the Quran. In particular, they should not be used as the basis for *tafsir* or for making religious or interpretive claims about the Quran. The resource is intended to support Quranic NLP research, facilitate downstream tasks, and improve the availability of Quranic resources with detailed morphosyntactic annotations.

As with many natural language processing tools, the morphosyntactic analyses and models presented in this work could potentially be misused. We en-

courage researchers and practitioners to use these resources only for ethical purposes and in accordance with applicable laws and ethical guidelines.

Large language models and AI-assisted writing tools were used solely to improve the clarity, grammar, and readability of the manuscript. All technical content, experimental design, analyses, and conclusions were developed and verified by the authors.

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A License

The original annotations created by the MAQAM authors are released under the Creative Commons Attribution 4.0 International License (CC BY 4.0), and the accompanying code is released under the MIT License. In Table 8, we list the license of the data and tools used in this work. All of them are used under their intended use.

Data/tool	License
Arabic Treebank: Part 1 v 4.1 (LDC2010T13)	LDC User Agreement for Non-Members
Arabic Treebank: Part 2 v 3.1 (LDC2011T09)	LDC User Agreement for Non-Members
Arabic Treebank: Part 3 v 3.2 (LDC2010T08)	LDC User Agreement for Non-Members
CAMeL Tools (Obeid et al., 2020)	MIT License
CAMeLBERT (Inoue et al., 2021)	MIT License
Character Edit Distance (CED) word alignment tool (Khalifa et al.)	MIT License.
CamelMorph (Khairallah et al., 2024)	MIT License
Tanzil Quran Text (Tanzil Project, 2021)	Creative Commons Attribution 3.0 (CC BY 3.0).
MASQA Data (Sawalha et al., 2025)	Creative Commons Attribution 4.0 license
Quranic Arabic Corpus (Dukes and Habash, 2010)	GNU General Public License
QuranMorph (Akra et al., 2025)	Creative Commons Attribution 4.0 license
MAQAM original annotations (ours)	Creative Commons Attribution 4.0 License
MAQAM source code (ours)	MIT License

Table 8: License of the data and tools.

B Per-Feature Synchronization Evaluation

Feature	All Words	INV Words	OOV Words
<i>prc3</i>	99.8	99.8	100.0
<i>prc2</i>	98.3	98.3	96.9
<i>prc1</i>	98.6	98.7	96.9
<i>prc0</i>	96.9	96.8	100.0
<i>per</i>	99.6	99.6	100.0
<i>asp</i>	99.6	99.6	98.4
<i>vox</i>	99.6	99.6	98.4
<i>mod</i>	99.1	99.2	93.8
<i>form_gen</i>	95.8	95.8	95.3
<i>gen</i>	95.4	95.4	96.9
<i>form_num</i>	98.8	98.9	95.3
<i>num</i>	98.3	98.4	95.3
<i>stt</i>	96.9	96.9	96.9
<i>cas</i>	96.0	96.0	98.4
<i>enc0</i>	99.4	99.5	93.8
<i>enc1</i>	100.0	100.0	96.9
Average	98.3	98.3	97.1

Table 9: Q-Test synchronization accuracies (%) for all, in-vocabulary (INV), and out-of-vocabulary (OOV) words.